

Technology, Inequality, Market Structure, And Labor-Market Institutions

Market-Level Incidence Of Technological Change

Special Topic 5: Technology, Innovation, and Labor – Week 5

Central question

- When technology raises productivity, who captures the gains?
- Do gains appear as wages, lower prices, rents, markups, or firm growth?
- Which workers move into better jobs, and which workers move into worse job-quality regimes?
- Week 5 discipline: market incidence is a labor question.

From firm adoption to market incidence

$$\Delta Y_m^L = \sum_f s_{fm} \Delta y_{fm}^L + \sum_f \Delta s_{fm} y_{fm}^L + \sum_j \Delta \omega_{jm} y_{jm}^L$$

- Within firms: productivity, wages, labor share, training, worker voice.
- Between firms: market share shifts toward different labor-share and wage-setting regimes.
- Between sectors: workers move across agriculture, manufacturing, formal services, informal services, and digital services.
- The same technology can raise output and worsen worker incidence.

Rent capture, labor share, and superstar firms

- Technology can create rents through patents, data, software, organizational capital, and scale.
- Superstar-firm logic: market share shifts toward high-productivity, low-labor-share firms.
- Aggregate labor share can fall even without a representative firm becoming less labor friendly.
- Labor questions: which workers share rents, which workers lose outside options, and which jobs expand?

Market power and institutions

Margin	Technology channel	Labor incidence
Employer concentration	growth of dominant adopters	weaker outside options
Intangible rents	data, patents, platforms, brands	uneven rent sharing
Platform rules	matching, ratings, scheduling	risk and control shift
Worker voice	unions, bargaining, transparency	incidence mediation

- Institutions do not change the engineering frontier, but they change who captures surplus.

Historical technology waves

Wave	Main labor margin	Persistence question
Mechanization	manual effort and factory scale	did displaced tasks return elsewhere?
Electrification	workflow and organization	how long did complements take?
Computerization	routine cognitive tasks	did polarization persist?
Robots	embodied production tasks	did local losses offset firm gains?
AI	prediction, language, coding, control	new tasks or deeper reallocation?

- AI is new in capabilities, but it still must be read through task reallocation, diffusion, complements, and institutions.

Structural change and job quality

- Agriculture to manufacturing to services is not only a productivity story.
- Mature economies face manufacturing-to-services transitions after automation and trade shocks.
- Employment reallocation is not the same as welfare-improving reallocation.
- Job-quality objects: wages, stability, voice, schedules, safety, formality, and mobility ladders.
- Formal services and informal services absorb workers in very different ways.

Global and Global South evidence

- Developing-country technology incidence often runs through diffusion, absorptive capacity, informality, and management.
- Labor-absorbing services can be a development path or a low-productivity absorption margin.
- Outsourcing and offshoring move tasks across countries before all jobs are automated.
- AI may complement scarce skilled labor in some settings and displace digital service tasks in others.
- High-income automation evidence is not automatically externally valid.

Frontier AI infrastructure questions

- Compute-intensive AI can shift electricity demand and data-center location.
- Data-center buildout affects construction, technical operations, local services, land use, and local fiscal capacity.
- Energy bottlenecks can change which firms and regions can adopt AI at scale.
- Outsourced digital services may expand through data work, localization, oversight, and AI-assisted knowledge services.
- The labor object is spatial incidence, not generic energy commentary.

Design family	Labor object
Market-share decomposition	within-firm versus between-firm labor-share change
Concentration accounting	product-market and labor-market power
Shift-share exposure	local incidence of technology shocks
Historical comparison	persistence, reversal, and complements
Structural-change decomposition	sectoral employment versus job quality
Linked firm-worker data	wage premia, training, exits, and mobility
Cross-country adoption surveys	diffusion and absorptive capacity
Energy and spatial data	AI infrastructure and local labor demand

- Reproduce: build a synthetic industry-firm panel with market share, labor share, productivity, and technology intensity.
- Diagnose: separate direct technology measurement from inferred technology, concentration, wage setting, and institutions.
- Transfer: move from superstar firms to structural change, Global South diffusion, outsourcing, and AI infrastructure.
- Output: compact tables on labor-share decomposition, incidence diagnosis, and global/frontier transfer scenarios.
- Goal: practice market-level labor incidence without confidential firm microdata.

- Week 5 aggregates technology from firms and workers into market-level incidence.
- Week 6 turns the whole course into research designs.
- A strong project names the technology margin, exposure measure, labor outcome, mediator, and counterfactual.

- Technology changes who captures rents, not only how much output is produced.
- Labor share, concentration, firm rents, and wage setting must be studied together.
- Historical waves help discipline AI claims without flattening what is new.
- Structural change should be evaluated by job quality and welfare, not employment movement alone.
- Global evidence changes the interpretation of technology shocks and opens infrastructure and outsourcing frontiers.